AME 5101 - Applied Linear Algebra

Assignment 1

Maximum marks: 10 Duration: 1 hour

Linear independence - A Worked Example.

Linear dependence and independence [CO2-PO3-BL4]

(a) **[5]** Distinguish between a linearly independent and a linearly dependent set of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$. Analyse why any set of vectors that contains the zero vector must be linearly dependent.

Answer. Consider the equation $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$. Taking all $c_i = 0$ always satisfies it, so the real question is whether some other choice of the c_i also works.

The set is *linearly independent* if the equation holds only for $c_1 = \cdots = c_k = 0$. The set is *linearly dependent* if some combination with coefficients not all zero gives $\mathbf{0}$.

Now suppose the set contains the zero vector, say $\mathbf{v}_1 = \mathbf{0}$. Take $c_1 = 1$ and all other $c_i = 0$. Then $1 \cdot \mathbf{0} + 0 \cdot \mathbf{v}_2 + \cdots + 0 \cdot \mathbf{v}_k = \mathbf{0}$, because every term is the zero vector. One coefficient is non-zero, so this is a non-trivial combination, and the set is dependent.

(b) **[5]** Let $\mathbf{v}_1 = (1,\ 2,\ 0)^\top$, $\mathbf{v}_2 = (0,\ 1,\ 1)^\top$ and $\mathbf{v}_3 = (2,\ 5,\ 1)^\top$. Examine whether $\mathbf{v}_3$ can be written as $a\mathbf{v}_1 + b\mathbf{v}_2$ by determining the numbers a and b. Deduce whether the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, and justify your answer.

Answer. **Goal.** We have to decide whether $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent. **Idea:** if $\mathbf{v}_3$ can be built from $\mathbf{v}_1$ and $\mathbf{v}_2$, then $\mathbf{v}_3$ is not adding anything new, so the set is dependent. Hence we try to find numbers a and b with $\mathbf{v}_3 = a\mathbf{v}_1 + b\mathbf{v}_2$.

Steps.

1. Put $\mathbf{v}_3 = a\mathbf{v}_1 + b\mathbf{v}_2$ and compare the three components. This gives three equations in the two unknowns a and b:

$$\text{first: } 1a + 0b = 2, \qquad \text{second: } 2a + 1b = 5, \qquad \text{third: } 0a + 1b = 1.$$

2. Solve two of the equations. The first gives $a = 2$ and the third gives $b = 1$.
3. Check these values in the remaining (second) equation, because a and b must satisfy all three. Here $2 \cdot 2 + 1 = 5$, which is true. So $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$.
4. Take $\mathbf{v}_3$ to the left side: $2\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$. The coefficients $(2,\ 1,\ -1)$ are not all zero, so a non-trivial combination gives $\mathbf{0}$. Therefore the set is linearly **dependent**.

Remark. If the check in step 3 had failed, then no such a, b would exist, and $\mathbf{v}_3$ would not be a combination of $\mathbf{v}_1, \mathbf{v}_2$. That alone would not prove independence, and we would have to examine the set further.

The Assignment Question.

Q1. [CO2-PO3-BL4] **10 Marks**

Answer the following question on the same lines as the worked example shown above.

(a) [5] Distinguish between a linearly independent and a linearly dependent set of vectors. Analyse why a set of two non-zero vectors $\{\mathbf{u}, \mathbf{w}\}$ is linearly dependent exactly when one of them is a multiple of the other.

(b) [5] Let $\mathbf{v}_1 = (1,\ 0,\ 2)^\top$, $\mathbf{v}_2 = (2,\ 1,\ 3)^\top$ and $\mathbf{v}_3 = (0,\ 1,\ -1)^\top$. Examine whether $\mathbf{v}_3$ can be written as $a\mathbf{v}_1 + b\mathbf{v}_2$ by determining the numbers a and b. Deduce whether the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, and justify your answer.